

Zcash Protocol Specification

Version v2026.7.0-191-g0fae78 [NU6.2]

Daira-Emma Hopwood[†]
Sean Bowe[‡] — Taylor Hornby[§] — Nathan Wilcox[◇]

September 15, 2026



Abstract. **Zcash** is an implementation of the *Decentralized Anonymous Payment scheme Zerocash*, with security fixes and improvements to performance and functionality. It bridges the existing transparent payment scheme used by **Bitcoin** with a *shielded* payment scheme secured by *zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs)*. It attempted to address the problem of mining centralization by use of the *Equihash* memory-hard *proof-of-work* algorithm.

This specification defines the **Zcash** consensus protocol at launch, and after each of the upgrades codenamed **Overwinter**, **Sapling**, **Blossom**, **Heartwood**, **Canopy**, **NU5**, **NU6**, **NU6.1**, and **NU6.2**. It is a work in progress. Protocol differences from **Zerocash** and **Bitcoin** are also explained.

Keywords: anonymity, applications, cryptographic protocols, electronic commerce and payment, financial privacy, proof of work, zero knowledge.

Contents	1
1 Introduction	8
1.1 Caution	9

[†] Zcash Open Development Lab

[‡] Project Tachyon

[§] Independent consultant

[◇] Shielded Labs

¹ Jubjub bird image credit: Peter Newell 1902; Daira-Emma Hopwood 2018.